

# DINOJAN VIVEKANANTHARASA

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linkedin.com/in/dinojan | github.com/Dinojan9901 | Portfolio

## PROFESSIONAL SUMMARY

A fresh Computer Engineering graduate focused on **applied AI systems and machine learning**, with one year of professional full-stack experience shipping production platforms. Experienced in building LLM and RAG pipelines, fine-tuning open models with QLoRA, and evaluating them rigorously against baselines.

## EDUCATION

### BSc.Eng (Honours) in Computer Engineering

Mar 2022 – Sep 2026

*Faculty of Engineering, University of Ruhuna, Galle, Sri Lanka*

- **Soft Skill Development** (2022–2024)

### GCE Advanced Level

Aug 2018

*V/Vavuniya Tamil Madhya Maha Vidyalayam*

## PROFESSIONAL EXPERIENCE

### Full Stack Developer

Dec 2025 – May 2026

*Yarl Ventures (PVT) Ltd, Jaffna*

- Built EDUS LMS  end to end (React 19, TypeScript, Node.js, MongoDB, Redis): 4-role RBAC, attendance, S3 homework uploads, Socket.IO chat, and PayHere payments.
- Continued owning MediMan  releases, working in Agile sprints with designers and senior developers.

### Full Stack Developer Intern

May 2025 – Nov 2025

*Yarl Ventures (PVT) Ltd, Jaffna*

- Built the MediMan admin panel, adding server-side pagination and field projection to cut API payload sizes.
- Integrated Nodemailer, Twilio SMS/OTP, and Firebase push; debugged production via AWS CloudWatch logs.

## TECHNICAL SKILLS

**Programming:** Python, JavaScript, TypeScript, Node.js, SQL, C++, Java

**AI/ML:** LLM Fine-tuning (QLoRA), RAG, Transfer Learning, CNN, TensorFlow, PyTorch, Scikit-learn

**Tools:** FastAPI, Streamlit, ChromaDB, Sentence-Transformers, OpenAI/Groq API, Ollama, Docker, AWS

## NON-TECH ABILITIES

**Skills:** Time Management, Goal-Oriented, Teamwork, Problem Solving

## PROJECTS

### TRACES – Post-Quantum Secure Vehicular Network Defense

Dec 2025 – Sep 2026

*Python, NS-3, Qwen2.5-7B (QLoRA), Hyperledger Fabric*

- Designed a three-stage LLM anomaly-detection cascade (QLoRA fine-tuned Qwen2.5-7B), validated on 200 vehicles in NS-3.
- Pairs post-quantum cryptography (ML-KEM-1024, Falcon-1024) with Hyperledger Fabric zero-trust governance.

### AI-Powered Financial Document Q&A System (RAG)

Jan 2026 – May 2026

*Python, FastAPI, Streamlit, ChromaDB, Groq Llama 3.3 70B*

- Grounds Llama 3.3 70B in 10-K, earnings, and SEC PDFs through ChromaDB retrieval (800-token chunks, MiniLM embeddings), answering with page-level citations.
- Evaluated on financial-qa-10K: Hit@1 retrieval of 1.000 and 84% gold-keyword recovery vs under 1% for an ungrounded LLM; deployed live with a free local fallback.

## AutoGradePro – AI-Powered Grading System

Jun 2024 – May 2025

*Next.js, TypeScript, Django, Ollama*

- Hybrid grader pairing rule-based checks with local LLM semantic matching (Ollama), so short answers are scored on meaning rather than exact keywords; published at Academic Sessions 2026.

## ECG Image Classification

Dec 2024 – Apr 2025

*Python, TensorFlow, CNN*

- Developed a CNN for ECG classification on MIT-BIH and PTB datasets using transfer learning, building the full pipeline from preprocessing to evaluation.

## Personal AI Assistant Web App

Apr 2025 – May 2025

*React.js, Node.js, OpenAI API*

- Built a full-stack conversational assistant with OpenAI integration, JWT auth, and persistent chat history.

## Room Occupancy Detection

Jun 2024 – Nov 2024

*Python, Scikit-learn*

- Built ML classifiers on multi-sensor IoT readings for smart-building occupancy detection, end to end.

## PUBLICATIONS

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G. S. J. Venujan, M. F. A. Munsif, **V. Dinojan**, T. Jegan, and G. G. N. Sandamali, “AutoGradePro: An Automated Grading System for Short-Answer Assessments Using Rule-Based Methods and Large Language Models,” in *Proceedings of Academic Sessions 2026*, University of Ruhuna, Sri Lanka, Mar. 2026, Abstract ABS-064. ISSN 2362-0412.

## VOLUNTEER EXPERIENCE

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### Demonstrator – ReXtro 2025 Silver Jubilee Engineering Exhibition

Dec 2025

*Software Engineering Zone, Faculty of Engineering, University of Ruhuna*

### Member

Apr 2024 – Present

*IESL, Galle*

### Member

Sep 2023 – Dec 2024

*IEEE Student Branch, Ruhuna, Galle*

### Treasurer

May 2021 – Apr 2024

*All University Students' Development Association, Vavuniya*

### Senior Prefect

Jul 2017 – Jul 2018

*V/V.T.M.M.V, Vavuniya*

## ACHIEVEMENTS

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Participated in **IEEE Xtreme 18.0 (2024)**, a global 24-hour algorithmic coding contest (3-member team)  
Competed in **IEEE INSL Southern Provincial Competition**, 30th October 2024

## REFEREES

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### • Dr. Kushan Sudheera

Senior Lecturer

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### • Dr. Nadeesha Sandamali

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